

Experiment - 3

Aim: Implementation of max-heap sort algorithm.

Hardware Required:

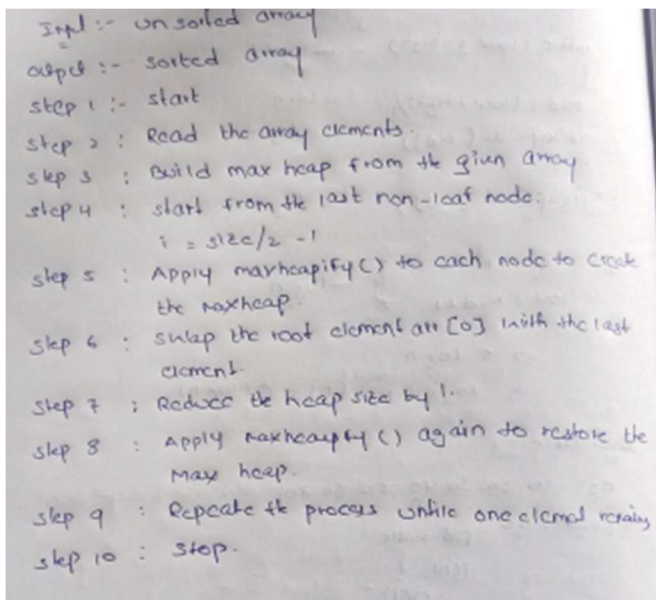
Mouse, keyboard.

Software Required:

code blocks.

Max-heap sort:

Algorithm:



Input :- unsorted array
output :- sorted array
step 1 :- start
step 2 :- Read the array elements.
step 3 :- Build max heap from the given array
step 4 :- start from the last non-leaf node:
 $i = \text{size}/2 - 1$
step 5 :- Apply maxheapify() to each node to create the maxheap.
step 6 :- swap the root element arr[0] with the last element.
step 7 :- Reduce the heap size by 1.
step 8 :- Apply maxheapify() again to restore the max heap.
step 9 :- Repeat the process until one element remains.
step 10 :- Stop.

Programm:

```
#include<iostream>

using namespace std;

class Heapsort
{
public:

    int arr[100],size;

    void arrayinput()
    {
        cout<<"enter size of array";
        cin>>size;

        cout<<"enter array elements";

        int i;
        for(i=0;i<size;i++)
        {
            cin>>arr[i];
        }
    }

    void printarray()
    {
```

```
int i;

for(i=0;i<size;i++)
{
    cout<<arr[i]<<" ";
}
}

void maxHeapify(int arr[],int size,int i)
{
    int largest=i;
    int l=(2*i)+1;
    int r=(2*i)+2;

    if(l<size && arr[l]>arr[largest])
    {
        largest=l;
    }

    if(r<size && arr[r]>arr[largest])
    {
        largest=r;
    }

    if(largest!=i)
    {
```

```
    swap(arr[largest],arr[i]);  
    maxHeapify(arr,size,largest);  
}  
}
```

```
void heaplogic()  
{  
    int i,temp;  
  
    for(i=size/2-1;i>=0;i--)  
    {  
        maxHeapify(arr,size,i);  
    }  
  
    for(i=size-1;i>0;i--)  
    {  
        temp=arr[0];  
        arr[0]=arr[i];  
        arr[i]=temp;  
  
        maxHeapify(arr,i,0);  
    }  
}  
};
```

```
int main()
{
    Heapsort hs;

    hs.arrayinput();

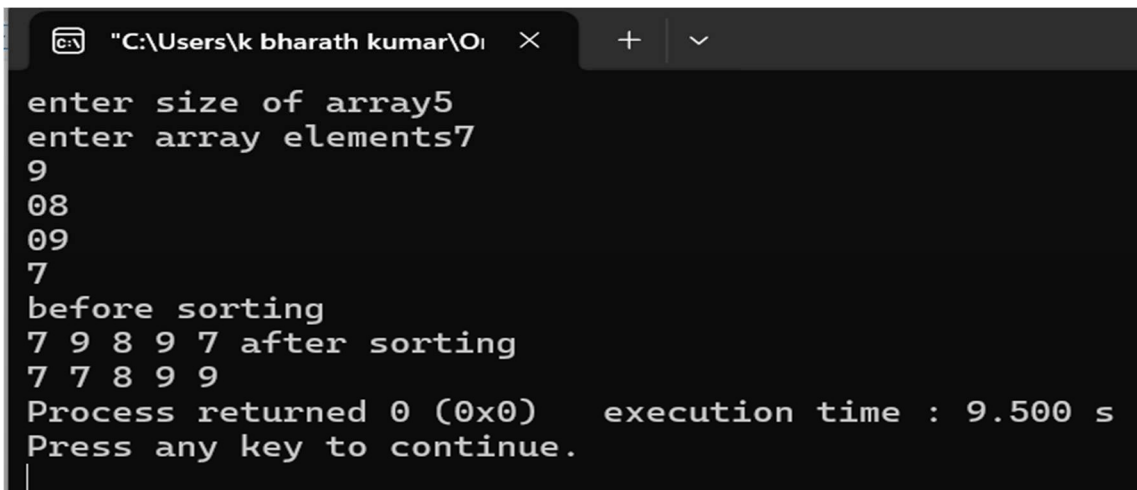
    cout<<"before sorting"<<endl;
    hs.printarray();

    hs.heaplogic();

    cout<<"after sorting"<<endl;
    hs.printarray();

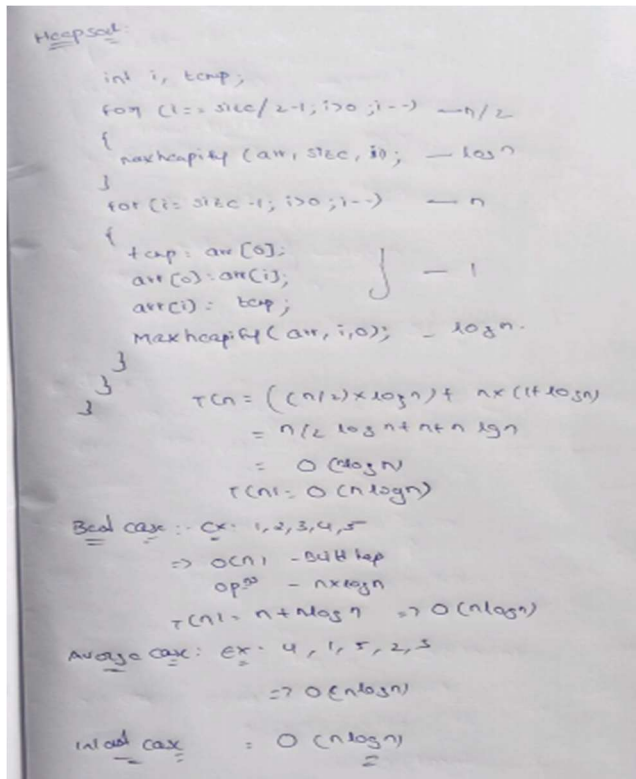
    return 0;
}
```

Output:



```
enter size of array5
enter array elements7
9
08
09
7
before sorting
7 9 8 9 7 after sorting
7 7 8 9 9
Process returned 0 (0x0) execution time : 9.500 s
Press any key to continue.
```

Time complexity:



conclusion:

As per the experiment to Implementation of max-heap sort algorithm, Output was verified successfully according to the Experiment.